Calibration frames

Print it, fold it, take it out with you.

Fragments of the Universe
bhapstar.com

The whole thing in one paragraph

Every frame you take holds the night sky plus a set of marks the camera and telescope put there on their own: dust shadows, darker corners, heat, and an electrical floor. Calibration frames are pictures of those marks with no sky in them, so the software can work out what to remove. Twenty minutes at the end of the night, and it is the difference between a clean picture and a blotchy one.

THE FOUR FRAMES, AND WHAT EACH ONE REMOVES

Frame	What it removes	How to take it	How many	How long it lasts
Dark	Heat and hot pixels	Cap on. Same exposure, gain and sensor temperature as your lights	20 to 30	Months, if the camera is cooled
Bias	The sensor's fixed offset	Cap on. The shortest exposure the camera will physically take	50 to 100	Until you change gain
Flat	Dust shadows and darker corners	Even light through the whole imaging train, with nothing moved or refocused	About 30	That session, that filter
Dark flat	The offset, out of the flats	Cap on. Same exposure as the flats you have just taken	Match the flats	With the flats

$$(\text{light} - \text{dark}) \div (\text{flat} - \text{bias})$$

Subtract what the sensor added.
Divide out what the optics took away. What is left is the sky.

AT THE SCOPE, IN THIS ORDER

1. **Lights**
   Shoot the target. Nothing on the rig moves, rotates or refocuses until the run is finished.
2. **Flats**
   Dawn sky, a flat panel, or a white T-shirt over the front with a tablet behind it. Same focus, same rotation, same filter. Set the exposure so the histogram peak sits near the middle.
3. **Dark flats**
   Cap on. Same exposure as the flats you just took, and the same number of them. Take these and you do not need biases at all.
4. **Darks**
   Cap on. Same exposure, gain and temperature as the lights. If the camera is not cooled, take them now while the air is still the temperature it was.
5. **Biases**
   Cap on. A thousandth of a second. They cost nothing and they keep for months, so they can be done indoors another day.
6. **Now pack up**
   Only once the flats are safely on the card is it safe to loosen, rotate or take anything apart.

THE RULES PEOPLE BREAK

Take flats before you take anything apart
The moment the camera rotates in the focuser, even slightly, the dust shadows move and that night's flats stop matching. This is the single most common mistake and it costs the whole set.

Do not touch focus
Dust shadows change size with focus position, so a flat taken at a different focus point removes the wrong shapes.

One set of flats per filter
Each filter carries its own dust and its own transmission. A quick-change filter drawer does not exempt you from re-shooting them.

Narrowband flats take longer than you expect
A 3nm filter blocks nearly everything, so a panel bright enough for an unfiltered flat gives you a nearly black one. Dim the panel and extend the exposure rather than the other way round.

Aim at something evenly lit
A cloud, a sunlit wall or a laptop screen with a gradient across it all produce a flat that makes the picture worse, because the software will faithfully divide out whatever unevenness you photographed.

WHAT IT COSTS TO SKIP THEM, AND WHERE IT HAPPENS

Skipping darks

Some extra noise and a scattering of coloured speckles. Survivable, and if you dithered during capture the stacking software rejects most of the hot pixels anyway.

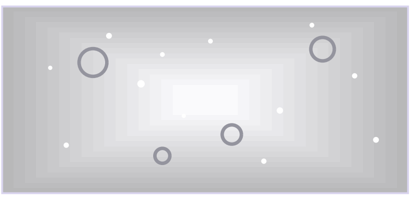
Skipping flats

This is the one that ruins pictures. Vignetting and dust are multiplied into the frame rather than added to it, so background extraction later cannot model them. The result looks nearly right and is subtly wrong everywhere.

Where it actually happens

Folders named lights, darks, flats and biases. Point Siril at the parent folder and run its one-shot-colour script: it stacks the calibration frames into masters, applies them, aligns everything and stacks the result, all unattended.

WHAT A FLAT FRAME ACTUALLY REMOVES



Your frame: corners lost, four dust shadows

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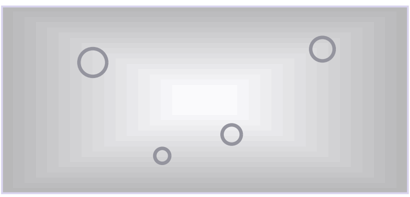

The flat: the same marks, on a blank field

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Corners back, marks gone, sky unchanged

The full guide, with examples

The reasoning behind each frame, what a flat actually removes, why gradient removal later cannot stand in for one, and the Siril workflow that puts it all together.

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